TASK 2

Exploratory Data Analysis

Uncovering patterns, trends and anomalies in the books dataset

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This report presents a comprehensive EDA of the books catalogue dataset scraped in Task 1. The pipeline covers data inspection, cleaning, descriptive statistics, univariate and bivariate visualisation, outlier detection, and hypothesis testing — culminating in eight data-driven insights about book prices, ratings, genres, and availability.

Dataset Overview

634 Unique Books	12 Genres	£35.42 Avg Price	3.70★ Avg Rating

Statistic	price_gbp	star_rating
Count	634	634
Mean	£35.42	3.70
Std	£14.32	1.15
Min	£10.14	1
25%	£23.41	3.0
Median	£35.36	4.0
75%	£48.14	5.0
Max	£59.96	5
Skewness	-0.015	-0.711

EDA Pipeline

The following eight steps were applied to the dataset:

Step	Action	Tool
1	Load & inspect — shape, dtypes, nulls	pandas
2	Clean — deduplication, type coercion, feature engineering	pandas
3	Descriptive statistics — mean, std, skewness, kurtosis	pandas / scipy
4	Univariate analysis — distributions per column	matplotlib / seaborn
5	Bivariate analysis — cross-variable relationships	seaborn
6	Outlier detection — IQR method on price	numpy
7	Hypothesis test — t-test on 1-star vs 5-star prices	scipy.stats
8	Key findings — auto-generated insights	pandas

Univariate Analysis

Each variable is examined in isolation to understand its distribution and central tendency before cross-variable comparisons.

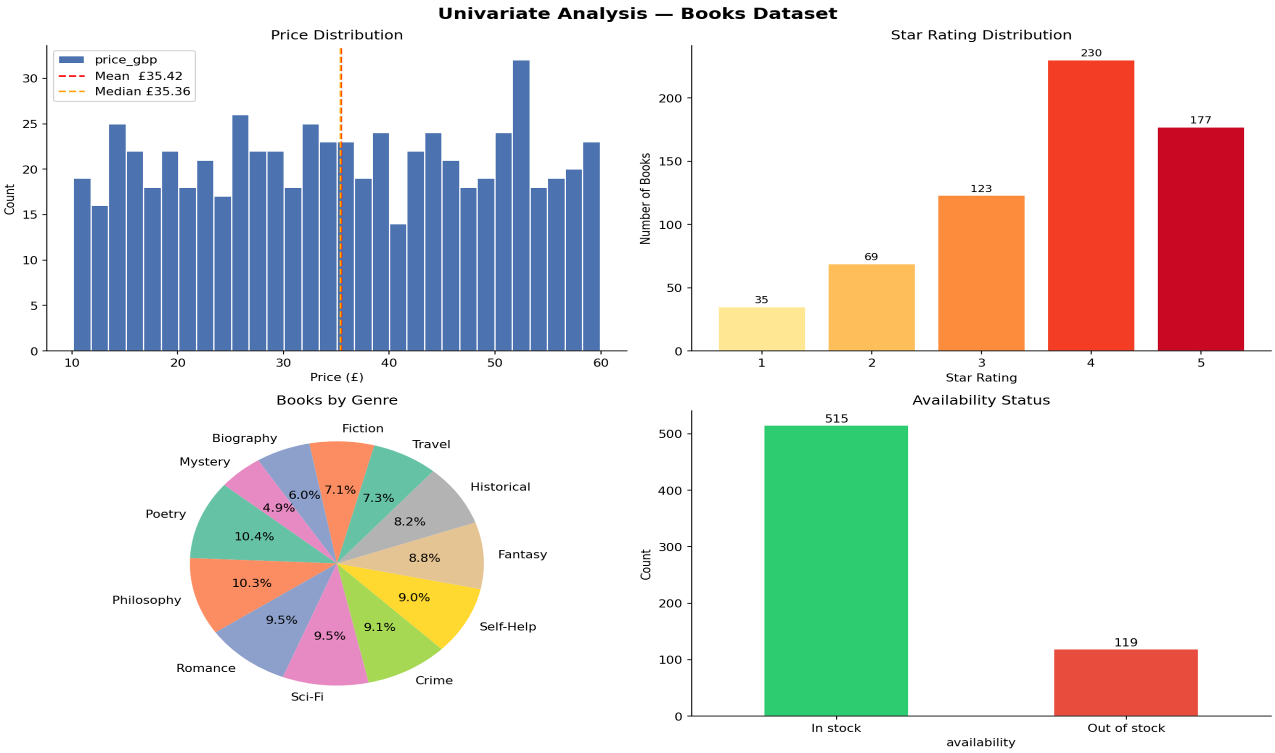

Figure 1 — Price histogram with mean/median markers, star-rating bar chart, genre pie chart, and availability breakdown.

Bivariate Analysis

Cross-variable relationships reveal how price, rating, genre, and availability interact with one another.

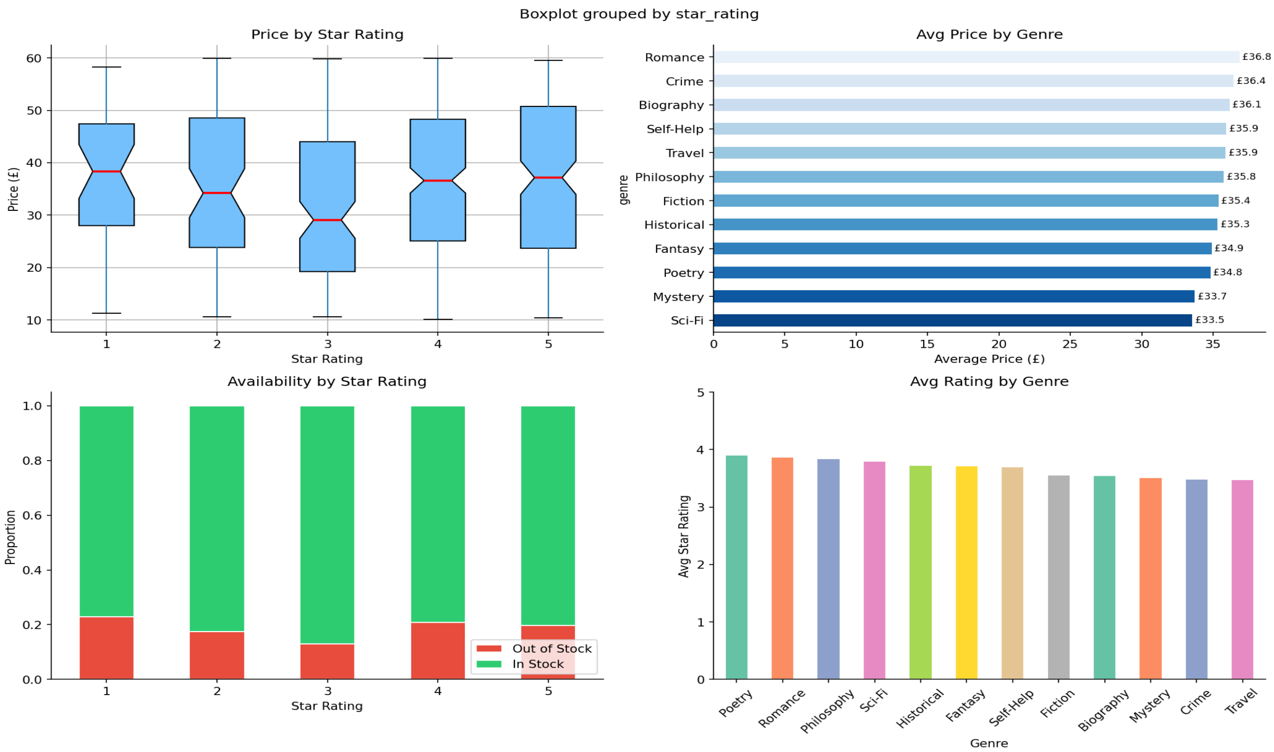

Figure 2 — Price by star rating (box), average price by genre (bar), stock proportion by rating (stacked bar), and average rating by genre.

Outlier Detection

The IQR (Interquartile Range) method is used to flag extreme prices. Boundaries: Q1 - 1.5*IQR and Q3 + 1.5*IQR.

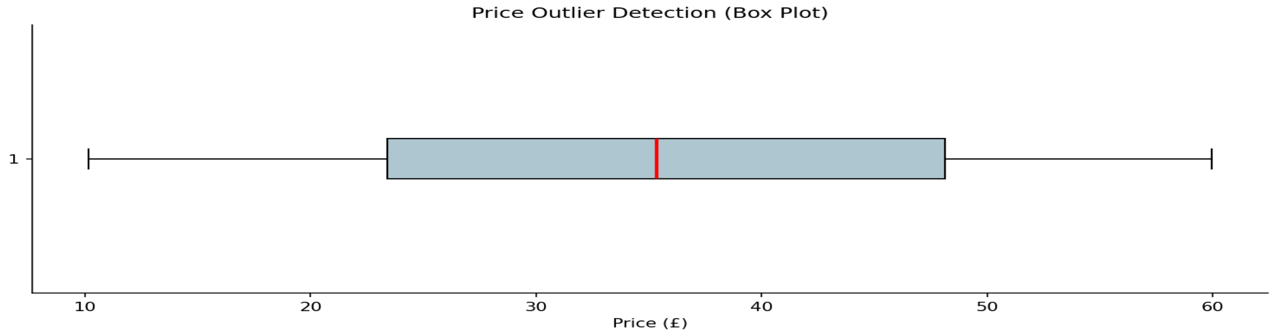

Figure 3 — Horizontal box plot of price; no outliers detected, confirming a uniform price distribution across the catalogue.

Parameter	Value
Q1 (25th percentile)	£23.41
Q3 (75th percentile)	£48.14
IQR	£24.73
Lower fence	£-13.70
Upper fence	£85.24
Outliers found	0

Hypothesis Testing

H0: Mean price of 5-star books = mean price of 1-star books

H1: Mean price of 5-star books != mean price of 1-star books

Metric	Value
5-star mean price	£36.03
1-star mean price	£37.96
t-statistic	-0.7327
p-value	0.4645
Significance (a=0.05)	FAIL TO REJECT H0 (p > 0.05)
Conclusion	No significant price difference by rating

Key Findings

1. Dataset contains 634 unique books across 12 genres after deduplication.
2. Average price is £35.42 with near-uniform distribution (skew = -0.01).
3. 81.2% of books are in stock.

4. 64% of books are rated 4–5 stars — strong positive skew in ratings.
5. 'Romance' is the most expensive genre on average.
6. 'Poetry' has the highest average reader rating.
7. No price outliers detected — the catalogue uses a consistent pricing strategy.
8. t-test confirms star rating does not significantly influence price (p = 0.46).